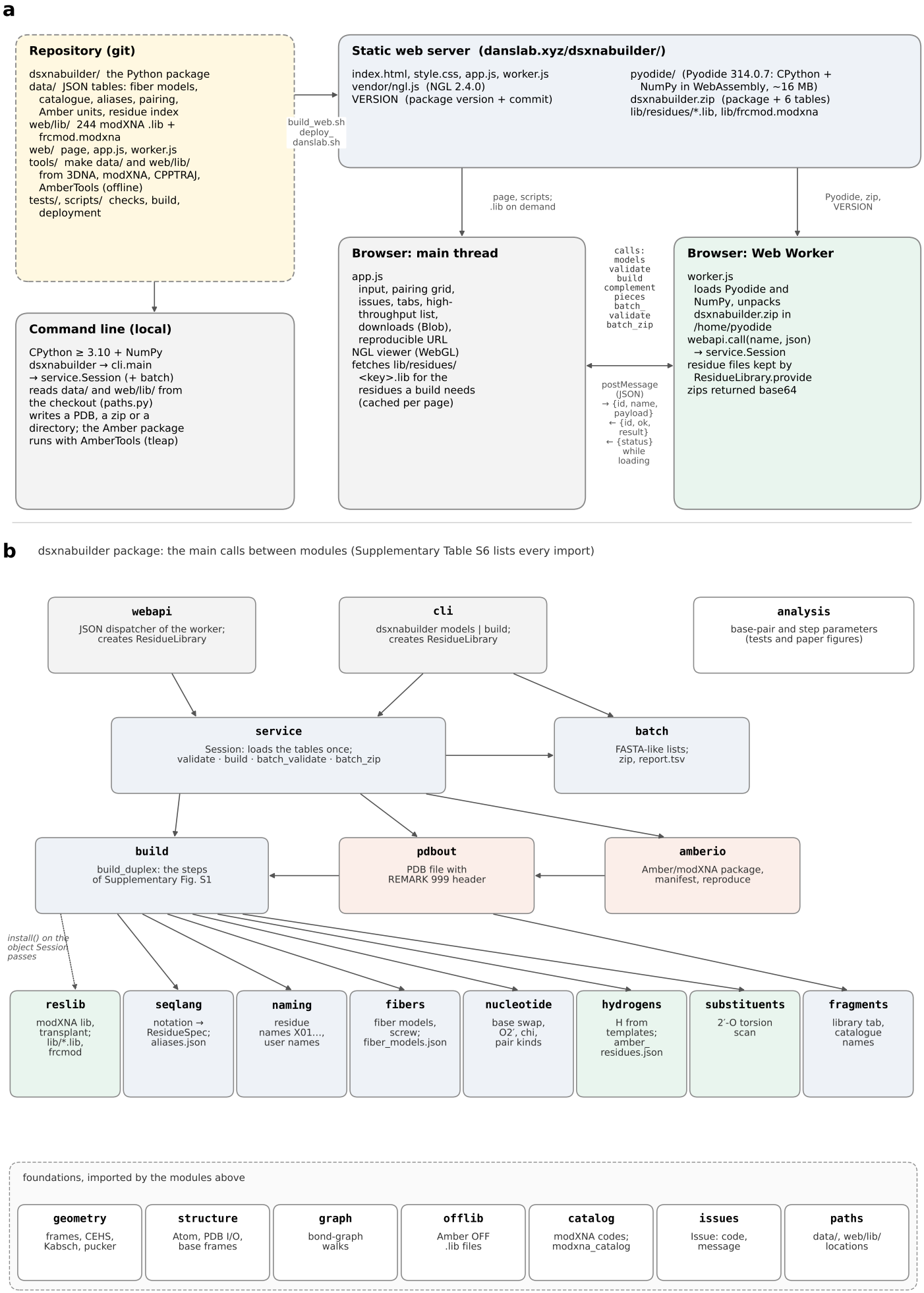

a
Repository (git)
dsxnabuilder/ the Python package
data/ JSON tables: fiber models, catalogue, aliases, pairing, Amber units, residue index
web/lib/ 244 modXNA .lib + frcmod.modxna
web/ page, app.js, worker.js
tools/ make data/ and web/lib/ from 3DNA, modXNA, CPPTRAJ, AmberTools (offline)
tests/, scripts/ checks, build, deployment
build_web.sh deploy_ danslab.sh
Static web server (danslab.xyz/dsxnabuilder/)
index.html, style.css, app.js, worker.js
vendor/ngl.js (NGL 2.4.0)
VERSION (package version + commit)
pyodide/ (Pyodide 314.0.7: CPython + NumPy in WebAssembly, ~16 MB)
dsxnabuilder.zip (package + 6 tables)
lib/residues/*.lib, lib/frcmod.modxna
page, scripts; .lib on demand
Pyodide, zip, VERSION
Command line (local)
CPython ≥ 3.10 + NumPy
dsxnabuilder → cli.main
→ service.Session (+ batch)
reads data/ and web/lib/ from the checkout (paths.py)
writes a PDB, a zip or a directory; the Amber package runs with AmberTools (tleap)
Browser: main thread
app.js
input, pairing grid, issues, tabs, high-throughput list, downloads (Blob), reproducible URL
NGL viewer (WebGL)
fetches lib/residues/ <key>.lib for the residues a build needs (cached per page)
calls: models validate build complement pieces batch_ validate batch_zip
postMessage (JSON) → {id, name, payload} ← {id, ok, result} ← {status} while loading
Browser: Web Worker
worker.js
loads Pyodide and NumPy, unpacks dsxnabuilder.zip in /home/pyodide
webapi.call(name, json)
→ service.Session
residue files kept by ResidueLibrary.provide
zips returned base64
b
dsxnabuilder package: the main calls between modules (Supplementary Table S6 lists every import)
webapi
JSON dispatcher of the worker; creates ResidueLibrary
cli
dsxnabuilder models | build; creates ResidueLibrary
analysis
base-pair and step parameters (tests and paper figures)
service
Session: loads the tables once; validate · build · batch_validate · batch_zip
batch
FASTA-like lists; zip, report.tsv
build
build_duplex: the steps of Supplementary Fig. S1
pdbout
PDB file with REMARK 999 header
amberio
Amber/modXNA package, manifest, reproduce
install() on the object Session passes
reslib
modXNA lib, transplant; lib/*.lib, frcmod
seqlang
notation → ResidueSpec; aliases.json
naming
residue names X01..., user names
fibers
fiber models, screw; fiber_models.json
nucleotide
base swap, O2′, chi, pair kinds
hydrogens
H from templates; amber_ residues.json
substituents
2′-O torsion scan
fragments
library tab, catalogue names
foundations, imported by the modules above
geometry
frames, CEHS, Kabsch, pucker
structure
Atom, PDB I/O, base frames
graph
bond-graph walks
offlib
Amber OFF .lib files
catalog
modXNA codes; modxna_catalog
issues
Issue: code, message
paths
data/, web/lib/ locations